

Project Executable Files

Date	1 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	No
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Project Workspace Directory Structure

```

opticrop/
├── data/
│   └── Crop_recommendation.csv    # Raw tabular agricultural profile records
├── models/
│   ├── crop_model.joblib         # Serialized Winner: Random Forest Classifier weights
│   └── scaler.joblib             # Serialized feature normalization weights
├── templates/
│   └── index.html                # Core application web UI structure layout
├── static/
│   ├── style.css                # Custom midnight-ui dashboard theme styles
│   └── script.js                # Asynchronous fetch client logic & client errors
├── notebook.ipynb               # Interactive prototyping & Exploratory Data Analysis
├── train.py                     # Production-grade evaluation & serialization pipeline
└── app.py                       # Flask Web Server API backend gateway
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	NIL
Login Credentials (Demo)	NIL
Platform / Hosting Provider	NIL
Repository Link	https://github.com/chEkshith/opti-crop
Demo Video Link	https://drive.google.com/file/d/1IknVrFXUy0yi7WW6KdEfTILa_7iIJmR5/view

Step 4: Run Instructions

1. Prerequisites

Ensure you have **Python 3.10+** and **Git** configured globally in your terminal environment.

2. Project Bootstrapping

Clone the workspace and navigate directly into the root directory:

```
git clone <your-repository-url>
cd opticrop
```

3. Dependencies Installation

Install the numeric computing, machine learning, and server libraries via **pip**:

```
pip install numpy pandas matplotlib seaborn scikit-learn joblib flask
```

Execution Guide

Step 1: Execute the Training Pipeline

Run the underlying production script to ingest data, execute structural validations, resolve internal parameters, and serialize the weights:

```
python train.py
```

Verification: Confirm that `models/crop_model.joblib` and `models/scaler.joblib` appear inside your workspace.

Step 2: Launch the Flask Server

Spin up the local microservices web instance gateway:

```
python app.py
```

Terminal will report: * Running on http://127.0.0.1:5000

Step 3: Access the Application Portal

Open your modern web browser and navigate directly to:

```
http://localhost:5000
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Port 5000 Conflict	<pre>app.run(port=5001, debug=True)</pre>
2	HTML Render Errors	Ensure index.html resides explicitly inside the templates/ subfolder. Flask utilizes Jinja2 lookups, which strictly fail if the layout template sits loose in the workspace root directory.